PARKTOWN BOYS' HIGH SCHOOL

ARISE

Department of Science

Preliminary exams

September 2023

Subject	Physical Sciences	**Marks**	150
Grade	12	**Time**	3 hours
Topic	Physics – Paper 1		

GENERAL INSTRUCTIONS

1. *Answer ALL the questions on your question paper.*
2. *The borrowing/lending of any material is FORBIDDEN.*
3. *Read the particular instructions for each section carefully (if applicable).*
4. *Answer all questions in the answer book provided.*
5. *Marks may be forfeited if ALL instructions are not followed.*
6. *This test consists of 21 pages which includes this cover page and the information sheets*
7. *Start each question on a new page*

Examiner	Mrs I Pelser
Internal Moderator	Dr L Whiting
External moderator	Mrs F Patel

Question 1 [20]

1.1 Consider the formula $W = F\Delta x\cos\theta$. How many scalar and how many vector quantities does this equation contain?

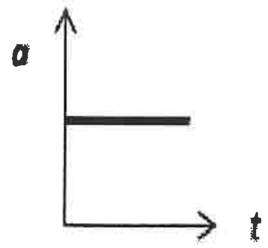
A one scalar and two vectors

B two scalars and one vector

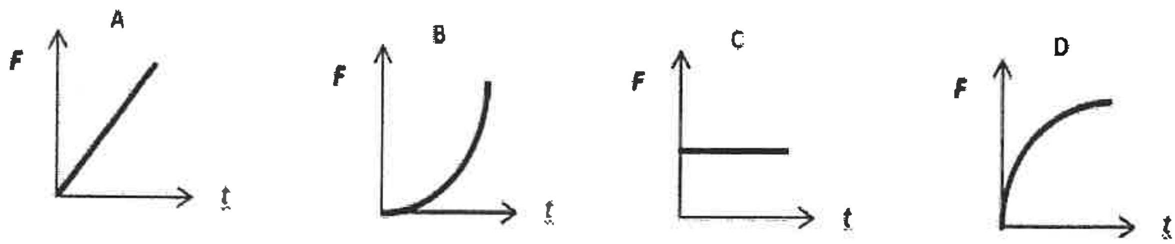
C three scalars

D three vectors

1.2 The acceleration-time graph for the motion of a certain body is shown in the graph below:



Which of the following graphs best illustrates the corresponding graph of the resultant force on the body versus time?



1.3 A ball is thrown vertically upward. The ball slows down and stops at its highest point. The acceleration at its highest point is:

A 0 m.s^{-2}

B 9.8 m.s^{-2} downward

C 9.8 m.s^{-2} upwards

D 0 m.s^{-1}

1.4 When the velocity of a moving object is doubled, the ...

A net work done by the object is doubled

B kinetic energy of the object is doubled

C potential energy of the object is doubled

D linear momentum of the object is doubled

1.5 The light from distant galaxies is shifted to longer wavelengths. This is called ______________ and occurs because __________

A red shift, the frequency of light decreases as it travels through space

B red shift, distant galaxies have a relative movement away from Earth

C blue shift, the wavelength of light from galaxies increases as it travels through space

D blue shift, some galaxies are approaching Earth

1.6 When a stone falls vertically which of the following statements is true? Ignore air friction

A. There is no change in the total energy

B. E_k is conserved

C E_k decreases and gravitational E_p increases

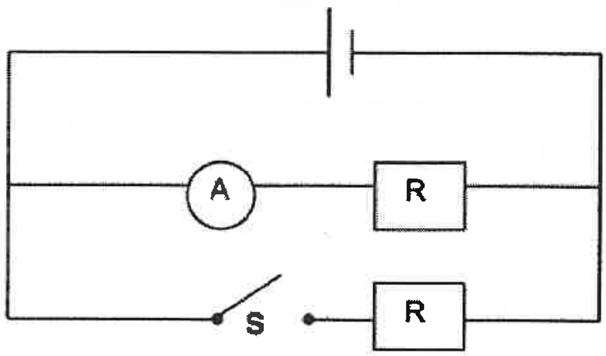
D. E_p is conserved

1.7 Two point charges, X and Y, produce a net electric field (E_{net}) at point P which is directed to the RIGHT, as shown in the diagram below.

Which ONE of the following combination of charges X and Y CANNOT have?

	X	Y
A	+	+
B	-	+
C	+	-
D	-	-

1.8 Two IDENTICAL resistors are connected in parallel as shown in the circuit diagram below. The internal resistance of the cell and resistance of the connecting wires can be ignored. When switch S is open, the reading of the ammeter is I.



When the switch S is closed, the reading of ammeter will be ...

A I

B 2I

C $\frac{I}{2}$

D 0

1.9 A multi plug enables us to plug many appliances in at the same time so that they can function simultaneously. If a multi plug adaptor connected in parallel is overloaded with too many appliances, a trip switch cuts off the electrical supply.
Which ONE of the following statements best explains the situation?

A The effective resistance increases and the current becomes too high

B Less current is required

C The effective resistance decreases, resulting in a larger current in the plug

D The current is divided among all the appliances and is not enough for them all to operate

1.10 The number of photoelectrons per unit time ejected from a metal surface depends on the ...

A intensity of the incident light

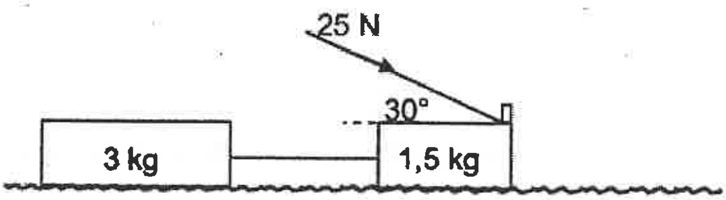
B frequency of the incident light

C wavelength of the incident light

D threshold frequency of the metal

Question 2 [17]

A learner constructs a push toy using two blocks with masses 1,5 kg and 3 kg respectively. The blocks are connected by a massless, inextensible cord. The learner then applies a force of 25 N at an angle of 30° to the 1,5 kg block by means of a light rigid rod, causing the toy to move across a flat, rough, horizontal surface, as shown in the diagram below. The coefficient of kinetic friction (μ_k) between the surface and each block is 0,15.



2.1 State Newton's Second Law of Motion in words. (2)

2.2 Calculate the magnitude of the kinetic frictional force acting on the 3 kg block. (3)

2.3 Draw a labelled free-body diagram showing ALL the forces acting on the 1,5 kg block. (3)

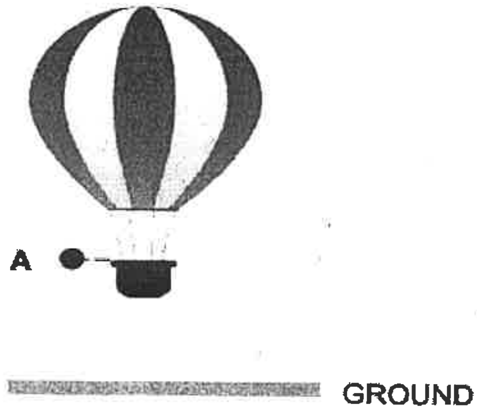
2.4 Calculate the magnitude of the:

2.4.1 Kinetic frictional force acting on the 1,5 kg block (4)

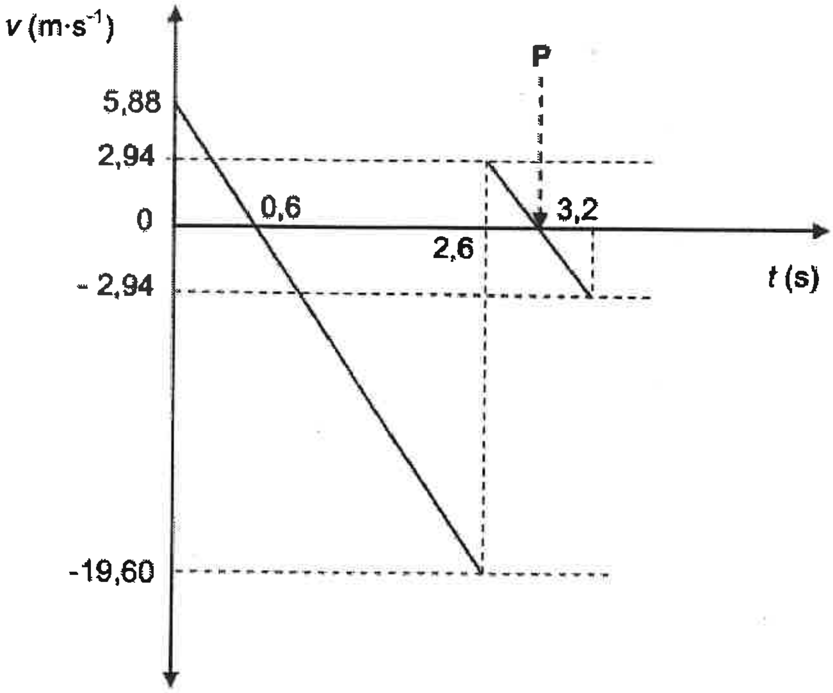
2.4.2 Tension in the cord connecting the two blocks (5)

Question 3 [15]

A hot air balloon rises vertically at a CONSTANT velocity. When the hot air balloon reaches point A few metres above the ground, a man in the balloon drops a ball which hits the ground and bounces. Ignore the effects of friction.



The velocity-time graph below represents the motion of the ball from the instant it is dropped until after it bounces for the first time. The time interval between bounces is ignored.



THE UPWARD DIRECTION IS TAKEN AS POSITIVE. USE INFORMATION FROM THE GRAPH TO ANSWER THE QUESTIONS THAT FOLLOW.

3.1 Define free fall (2)

3.2 Write down the magnitude of the velocity of the hot air balloon. (1)

3.3 Calculate the height above the ground from which the ball was dropped. (3)

Calculate the:

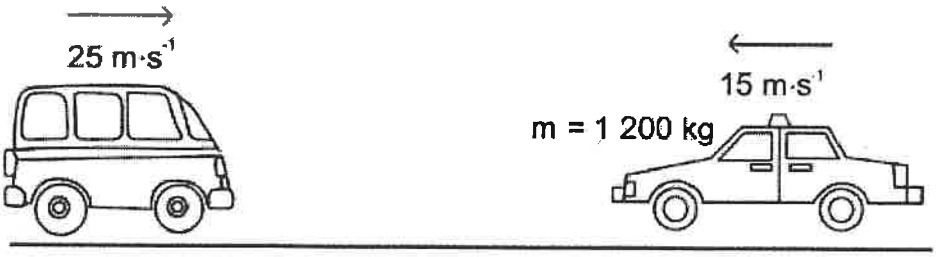
3.4 Time at the point P indicated on the graph (2)

3.5 Maximum height the ball reaches after the first bounce (3)

3.6 Distance between the ball and hot air balloon when the ball is at its maximum height after the first bounce (4)

Question 4 [11]

The average mass of a minibus taxi on South African roads is 1 500 kg. The law states that the combined mass of all the passengers in a minibus taxi and the taxi itself should not exceed 3500 kg.



A minibus taxi with an unknown number of passengers travels at 25 $m.s^{-1}$ when it collides with a car with a mass of 1 200 kg (passengers included), travelling at 15 $m.s^{-1}$ in the opposite direction, as shown. During the collision the vehicles stick together and travel at 14 $m.s^{-1}$ immediately after the collision in the direction of the original motion of the taxi.

4.1 State the law of conservation of momentum. (2)

4.2 Ignore friction. Use momentum principles to determine whether the minibus taxi was overloaded, that is, above the legal combined mass of 3 500 kg. (5)

4.3 Calculate the impulse on the car. (4)

Question 5 [12]

A 5 kg block is released from rest from a vertical height of 5 m and slides down a frictionless incline to point P as shown in the diagram below. It then moves along a frictionless horizontal portion PQ and finally moves up a second rough inclined plane. It comes to a stop at point R which is 3 m above the horizontal.

The frictional force, which is a non-conservative force, between the surface and the block is 18 N.

5.1 State the law of conservation of mechanical energy (2)

5.2 Using ENERGY PRINCIPLES only, calculate the speed of the block at point P. (3)

5.3 Explain why the kinetic energy at point P is the same as that at point Q. (2)

5.4 Calculate the angle (θ) of the slope QR. (5)

Question 6 [10]

An ambulance is moving towards a stationary listener at a constant speed of 30 m·s^{-1}. The siren of the ambulance emits sound waves with a wavelength of 0,28 m. The speed of sound in air as 340 m·s^{-1}.

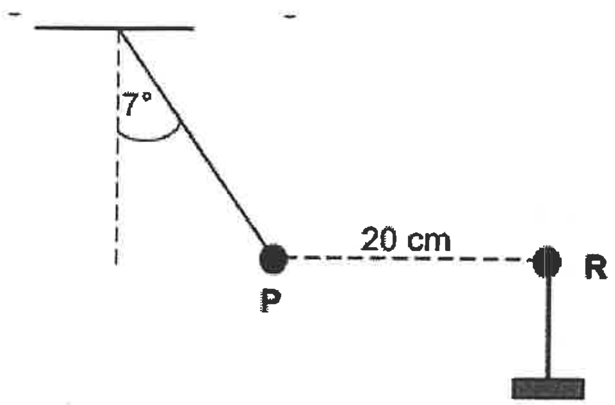
6.1 State the Doppler effect in words. (2)

6.2 Calculate the frequency of the sound waves emitted by the siren as heard by the ambulance driver. (3)

6.3 Calculate the frequency of the sound waves emitted by the siren as heard by the listener. (5)

Question 7 **[15]**

7.1 A very small graphite-coated sphere P is rubbed with a cloth. It is found that the sphere acquires a charge of + 0,5 μC.



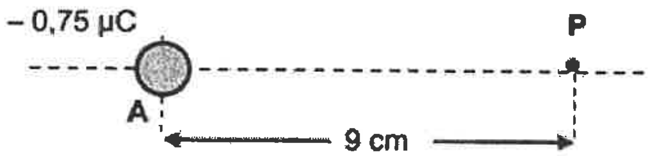
The charged sphere P is suspended from a light, inextensible string. Another sphere, R, with a charge of – 0,9 μC, supported on an insulated stand, is brought close to sphere P. As a result, sphere P moves to a position where it is 20 cm from sphere R, as shown below. The system is in equilibrium and the angle between the string and the vertical is 7°.

7.1.1 Draw a labelled free-body diagram showing ALL the forces acting on sphere P. (3)

7.1.2 State Coulomb's law in words. (2)

7.1.3 Calculate the magnitude of the tension in the string. (5)

7.2 A charged sphere, A, carries a charge of – 0,75 μC.



7.2.1 Draw a diagram showing the electric field lines surrounding sphere A. (2)

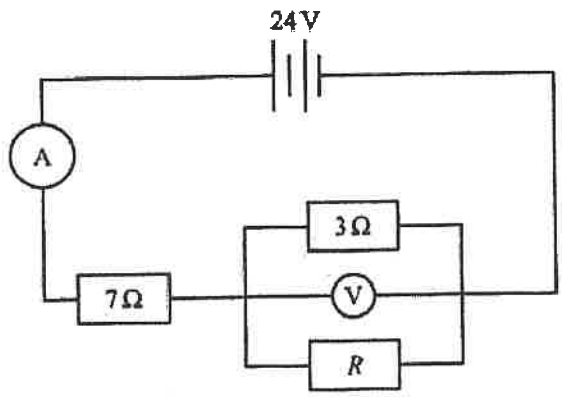
Point P is located 9 cm to the right of sphere A.

7.2.2 Calculate the magnitude of the net electric field at point P. (3)

Question 8 [25]

8.1 Three resistors are connected to a 24 V battery of negligible internal resistance as shown in the diagram. The battery supplies 60 W of power to the circuit.

The value of the resistance of resistor R is unknown.



8.1.1 Calculate the reading on the ammeter (3)

8.1.2 State Ohm's Law (2)

8.1.3 Calculate the potential difference across the 7Ω resistor (3)

8.1.4 Calculate the voltmeter reading (2)

8.1.5 Calculate the value of resistor R (5)

8.2 A set of students realize that the potential difference across the terminal of the battery, as well as the current change as the resistance in a circuit changes.

The students decide to experiment and changed the external resistance of a circuit while recording the following set of data.

Current (A)	Terminal potential difference (V)
0,15	11,75
0,35	11,10
0,75	10,23
1,15	9,15

8.2.1 Plot a graph of terminal potential difference vs current on the graph paper provided on your question paper.
Detach this graph paper and place it inside your answer book. Ensure your name clearly is written on this answer sheet (4)

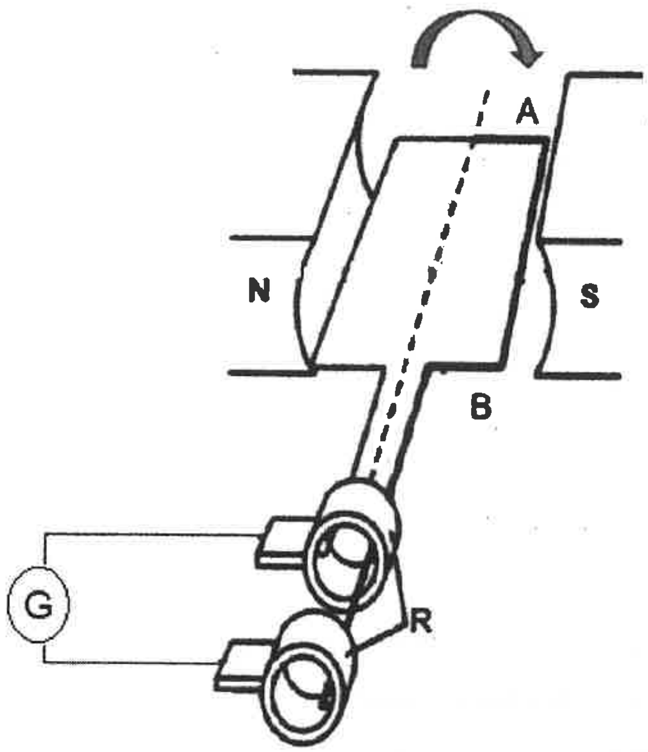
8.2.2 Calculate the gradient of the graph. Show all steps (3)

8.2.3 Use your answer in the previous question to determine the internal resistance of the battery (2)

8.2.4 State the value of the emf of the battery (1)

Question 9 [15]

The diagram given below is a schematic representation of an AC generator.



Write down the:

9.1.1 Name of the part labelled 'R'. (1)

9.1.2 Function of part labelled 'R'. (2)

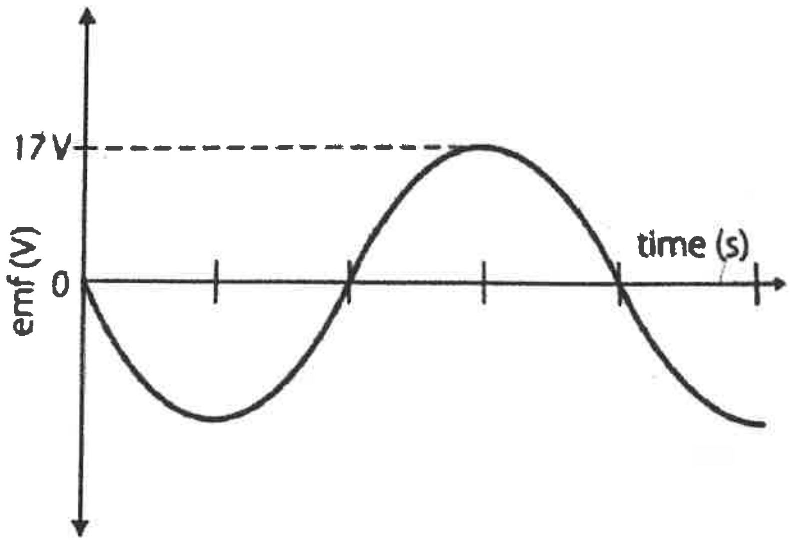
9.2 State the principle on which an AC generator works. (1)

9.3 If the armature is rotated in the direction as indicated in the diagram, which way will the current flow in the armature? Choose from A to B or B to A (1)

9.4 What effect does changing the polarity of the magnets have on the output voltage? Choose from INCREASES, DECREASES OR REMAINS THE SAME (1)

9.5 What is the position of the coil, relative to the magnetic field, when the output voltage is at a minimum? (2)

9.6 The graph shows the potential delivered by an AC source to a light bulb which is dissipating 15 W.

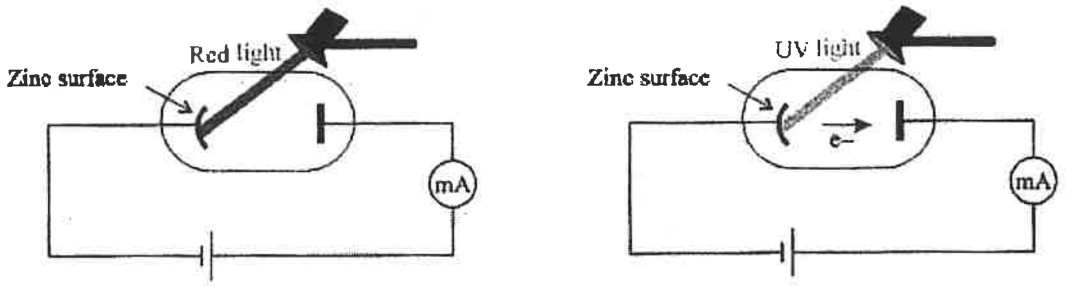


9.6.1 What DC voltage would have to be supplied to keep the bulb burning at the same brightness? (3)

9.6.2 What is the peak current through the bulb while it is connected to the AC source? (4)

Question 10 [10]

The diagram below shows an experiment where red light and UV light are shone onto a negatively charged plate of zinc metal



When red light is shone onto the negatively charged zinc plate the milliammeter reads 0 mA. When UV light is shone on the same plate, electrons are emitted and a reading is seen on the milliammeter.

10.1 Name the phenomenon described above (1)

10,2 What is the significance of this effect? (1)

10.3 Explain why red light has no effect but the UV light causes electrons to be emitted (2)

The work function of zinc is $6,928 \times 10^{-19}$ J.

10.4 Calculate the minimum kinetic energy of an electron ejected from the zinc metal by a photon of wavelength of 100 nm (4)

10.5 The intensity of both lights is increased:

For each experiment, state what would happen to the reading on the milliammeter. State only INCREASE, DECREASE or REMAIN THE SAME (2)

TOTAL: 150

DATA FOR PHYSICAL SCIENCES GRADE 12
PAPER 1 (PHYSICS)

GEGEWENS VIR FISIESE WETENSKAPPE GRAAD 12
VRAESTEL 1 (FISIKA)

TABLE 1: PHYSICAL CONSTANTS/*TABEL 1: FISIESE KONSTANTES*

NAME/*NAAM*	SYMBOL/*SIMBOOL*	VALUE/*WAARDE*
Acceleration due to gravity *Swaartekragversnelling*	g	$9,8\ m\cdot s^{-2}$
Universal gravitational constant *Universele gravitasiekonstant*	G	$6,67 \times 10^{-11}\ N\cdot m^2\cdot kg^{-2}$
Speed of light in a vacuum *Spoed van lig in 'n vakuum*	c	$3,0 \times 10^{8}\ m\cdot s^{-1}$
Planck's constant *Planck se konstante*	h	$6,63 \times 10^{-34}\ J\cdot s$
Coulomb's constant *Coulomb se konstante*	k	$9,0 \times 10^{9}\ N\cdot m^2\cdot C^{-2}$
Charge on electron *Lading op elektron*	e	$1,6 \times 10^{-19}\ C$
Electron mass *Elektronmassa*	m_e	$9,11 \times 10^{-31}\ kg$
Mass of the Earth *Massa van die Aarde*	M	$5,98 \times 10^{24}\ kg$
Radius of the Earth *Radius van die Aarde*	R_E	$6,38 \times 10^{6}\ m$

TABLE 2: FORMULAE/*TABEL 2: FORMULES*

MOTION/*BEWEGING*

$v_f = v_i + a\Delta t$	$\Delta x = v_i\Delta t + \frac{1}{2}a\Delta t^2$ or/*of* $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$
$v_f^2 = v_i^2 + 2a\Delta x$ or/*of* $v_f^2 = v_i^2 + 2a\Delta y$	$\Delta x = \left(\frac{v_i + v_f}{2}\right)\Delta t$ or/*of* $\Delta y = \left(\frac{v_i + v_f}{2}\right)\Delta t$

FORCE/*KRAG*

$F_{net} = ma$	$p = mv$
$f_s^{max} = \mu_s N$	$f_k = \mu_k N$
$F_{net}\Delta t = \Delta p$ $\Delta p = mv_f - mv_i$	$w = mg$
$F = G\frac{m_1m_2}{d^2}$ or/*of* $F = G\frac{m_1m_2}{r^2}$	$g = G\frac{M}{d^2}$ or/*of* $g = G\frac{M}{r^2}$

WORK, ENERGY AND POWER/*ARBEID, ENERGIE EN DRYWING*

$W = F\Delta x\cos\theta$	$U = mgh$ or/*of* $E_P = mgh$
$K = \frac{1}{2}mv^2$ or/*of* $E_k = \frac{1}{2}mv^2$	$W_{net} = \Delta K$ or/*of* $W_{net} = \Delta E_k$ $\Delta K = K_f - K_i$ or/*of* $\Delta E_k = E_{kf} - E_{ki}$
$W_{nc} = \Delta K + \Delta U$ or/*of* $W_{nc} = \Delta E_k + \Delta E_p$	$P = \frac{W}{\Delta t}$
$P_{ave} = Fv_{ave}$ / $P_{gemid} = Fv_{gemid}$	

WAVES, SOUND AND LIGHT/*GOLWE, KLANK EN LIG*

$v = f\lambda$	$T = \frac{1}{f}$
$f_L = \frac{v \pm v_L}{v \pm v_s}f_s$ $f_L = \frac{v \pm v_L}{v \pm v_b}f_b$	$E = hf$ or/*of* $E = h\frac{c}{\lambda}$
$E = W_o + E_{k(max)}$ or/*of* $E = W_o + K_{max}$ where/*waar* $E = hf$ and/*en* $W_0 = hf_0$ and/*en* $E_{k(max)} = \frac{1}{2}mv_{max}^2$ or/*of* $K_{max} = \frac{1}{2}mv_{max}^2$	

ELECTROSTATICS/*ELEKTROSTATIKA*

$F=\frac{kQ_1Q_2}{r^2}$	$E=\frac{kQ}{r^2}$
$V=\frac{W}{q}$	$E=\frac{F}{q}$
$n=\frac{Q}{e}$ or/*of* $n=\frac{Q}{q_e}$	

ELECTRIC CIRCUITS/*ELEKTRIESE STROOMBANE*

$R=\frac{V}{I}$	emf (ε) = $I(R + r)$ emk (ε) = $I(R + r)$
$R_s=R_1+R_2+\ldots$ $\frac{1}{R_p}=\frac{1}{R_1}+\frac{1}{R_2}+\ldots$	$q=I\Delta t$
$W = Vq$ $W = VI\Delta t$ $W = I^2R\Delta t$ $W = \frac{V^2\Delta t}{R}$	$P=\frac{W}{\Delta t}$ $P = VI$ $P = I^2R$ $P=\frac{V^2}{R}$

ALTERNATING CURRENT/*WISSELSTROOM*

$I_{rms}=\frac{I_{max}}{\sqrt{2}}$ / $I_{wgk}=\frac{I_{maks}}{\sqrt{2}}$ $V_{rms}=\frac{V_{max}}{\sqrt{2}}$ / $V_{wgk}=\frac{V_{maks}}{\sqrt{2}}$	$P_{ave}=V_{rms}I_{rms}$ / $P_{gemiddeld}=V_{wgk}I_{wgk}$ $P_{ave}=I_{rms}^2R$ / $P_{gemiddeld}=I_{wgk}^2R$ $P_{ave}=\frac{V_{rms}^2}{R}$ / $P_{gemiddeld}=\frac{V_{wgk}^2}{R}$